SYLLABUS : ELASTICITY & VISCOSITY + SURFACE TENSION

1. Insects are able to run on the surface of water because :

(A) insects have less weight

(B) insects swim on water

(C) of the Archimede's upthrust

(D) surface tension makes the surface behave as elastic membrane.

2. The radii of the two columns is U-tube are r_1 and r_2. When a liquid of density ρ (angle of contact is 0°) is filled in it, the level difference of liquid in two arms is h. The surface tension of liquid is: (g = acceleration due to gravity) :

(A) $\frac{\rho g h r_1 r_2}{2(r_2 - r_1)}$ (B) $\frac{\rho g h (r_2 - r_1)}{2 r_1 r_2}$ (C) $\frac{2(r_2 - r_1)}{\rho g h r_1 r_2}$ (D) $\frac{\rho g h}{2(r_2 - r_1)}$

3. In a surface tension experiment with a capillary tube water rises upto 0.1 m. If the same experiment is repeated in an artificial satellite, which is revolving around the earth ; water will rise in the capillary tube upto a height of :

(A) 0.1 m (B) 0.2 m (C) 0.98 m (D) full length of tube

4. A thin metal disc of radius r floats on water surface and bends the surface downwards along the perimeter making an angle θ with vertical edge of the disc. If the disc displaces a weight of water W and surface tension of water is T, then the weight of metal disc is :

(A) 2π rT + W (B) 2π rT cosθ – W (C) 2π rT cosθ + W (D) W – 2π rT cosθ

5. The surface tension of a liquid is 5 Newton per metre. If a film is held on a ring of area 0.02 metres^2, its surface energy is about :

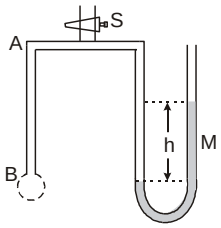
(A) 5×10^{-2} J (B) 2.5×10^{-2} J (C) 2×10^{-1} J (D) 3×10^{-1} J

6. Water rises in a capillary tube to a height h. it will rise to a height more than h

(A) on the surface of sun (B) in a lift moving down with an acceleration

(C) at the poles (D) in a lift moving up with an acceleration.

7. A tube of fine bore AB is connected to a manometer M as shown. The stop cock S controls the flow of air. AB is dipped into a liquid whose surface tension is σ. On opening the stop cock for a while, a bubble is formed at B and the manometer level is recorded, showing a difference h in the levels in the two arms. if ρ be the density of manometer liquid and r the radius of curvature of the bubble, then the surface tension σ of the liquid is given by

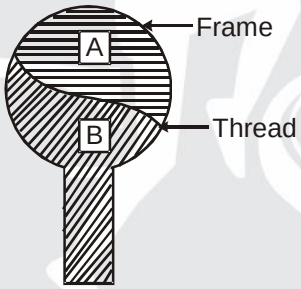


(A) ρhrg (B) $2\rho hgr$ (C) $4\rho hrg$ (D) $\frac{rh\rho g}{4}$

8. Two parallel glass plates are dipped partly in the liquid of density 'd'. keeping them vertical. If the distance between the plates is 'x', Surface tension for liquid is T & angle of contact is θ then rise of liquid between the plates due to capillary will be :

(A) $\frac{T\cos\theta}{x\ \ d}$ (B) $\frac{2T\cos\theta}{xdg}$ (C) $\frac{2T}{xdg\cos\theta}$ (D) $\frac{T\cos\theta}{xdg}$

9. A thread is tied slightly loose to a wire frame as shown in the figure. And the frame is dipped into a soap solution and taken out. The frame is completely covered with the film. When the portion A is punctured with a pin, the thread :



(A) becomes convex towards A

(B) becomes concave towards A

(C) remains in the initial position

(D) either (A) or (B) depending on size of A w.r.t. B

10. A cylinder with a movable piston contains air under a pressure p_1 and a soap bubble of radius 'r'. The pressure p_2 to which the air should be compressed by slowly pushing the piston into the cylinder for the soap bubble to reduce its size by half will be : (The surface tension is σ, and the temperature T is maintained constant)

(A) $\left[8p_1+\frac{24\sigma}{r}\right]$ (B) $\left[4p_1+\frac{24\sigma}{r}\right]$ (C) $\left[2p_1+\frac{24\sigma}{r}\right]$ (D) $\left[2p_1+\frac{12\sigma}{r}\right]$

11. A water drop is divided into 8 equal droplets. The pressure difference between the inner and outer side of the big drop will be :

(A) same as for smaller droplet (B) 1/2 of that for smaller droplet

(C) 1/4 of that for smaller droplet (D) twice that for smaller droplet

12. A vessel whose bottom has round holes with a diameter of d = 0.1 mm is filled with water. The maximum height of the water level h at which the water does not flow out, will be : (The water does not wet the bottom of the vessel). [S.T of water = 70 dyn/cm]

(A) h = 24.0 cm (B) h = 25.0 cm (C) h = 26.0 cm (D) h = 28.0 cm

13. An air bubble of radius r in water is at a depth h below the water surface at some instant. If P is atmospheric pressure, d and T are density and surface tension of water respectively, the pressure inside the bubble will be :

(A) $P + h\,dg - \frac{4T}{r}$ (B) $P + h\,dg + \frac{2T}{r}$ (C) $P + h\,dg - \frac{2T}{r}$ (D) $P + h\,dg + \frac{4T}{r}$

14. The work done to get n smaller equal size spherical drops from a bigger size spherical drop of water is proportional to :

(A) $\left(\frac{1}{n^{2/3}}\right) - 1$ (B) $\left(\frac{1}{n^{1/3}}\right) - 1$ (C) $n^{1/3} - 1$ (D) $n^{4/3} - 1$

15. Two unequal soap bubbles are formed one on each side of a tube closed in the middle by a tap. What happens when the tap is opened to put the two bubbles in communication ?

(A) No air passes in any direction as the pressures are the same on two sides of the tap

(B) Larger bubble shrinks and smaller bubble increases in size till they become equal in size

(C) Smaller bubble gradually collapses and the bigger one increases in size

(D) None of the above

16. When charge is given to a soap bubble, it shows :

(A) a decrease in size

(B) no change in size

(C) an increase in size

(D) sometimes an increase and sometimes a decreases in size

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17. A soap bubble in vacuum has a radius of 3 cm and another soap bubble in vacuum has a radius of 4 cm. If the two bubbles coalesce under isothermal conditions then the radius of the new bubble is :

(A) 2.3 cm (B) 4.5 cm (C) 5 cm (D) 7 cm

18. At critical temperature, the surface tension of a liquid :

(A) is zero (B) is infinity

(C) is same as that at any other temperature (D) cannot be determined

19. A soap bubble of radius r_1 is placed on another soap bubble of radius $r_2 (r_1 < r_2)$. The radius R of the soapy film separating the two bubbles is :

(A) $r_1 + r_2$ (B) $\sqrt{r_1^2 + r_2^2}$ (C) $(r_1^3 + r_2^3)$ (D) $\frac{r_2 r_1}{r_2 - r_1}$

20. A capillary tube of radius R is immersed in water and water rises in it to a height H. Mass of water in capillary tube is M. If the radius of the tube is doubled, mass of water that will rise in capillary tube will be

(A) 2M (B) M (C) $\frac{M}{2}$ (D) 4M

21. Work done in increasing the size of a soap bubble from a radius of 3 cm to 5 cm is nearly. (Surface tension of soap solution = 0.03 Nm^{-1})

(A) 4π mJ (B) 0.2π mJ (C) 2π mJ (D) 0.4π mJ

22. Two mercury drops (each of radius 'r') merge to from bigger drop. The surface energy of the bigger drop is, if T is the surface tension :

(A) $4\pi r^2 T$ (B) $2\pi r^2 T$ (C) $2^{8/3}\pi r^2 T$ (D) $2^{5/3}\pi r^2 T$

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23. Assume that a drop of liquid evaporates by decreasing in its surface energy, so that its temperature remains unchanged. What should be the minimum radius of the drop for this to be possible? The surface tension is T, density of liquid is ρ and L is its latent heat of vaporization.

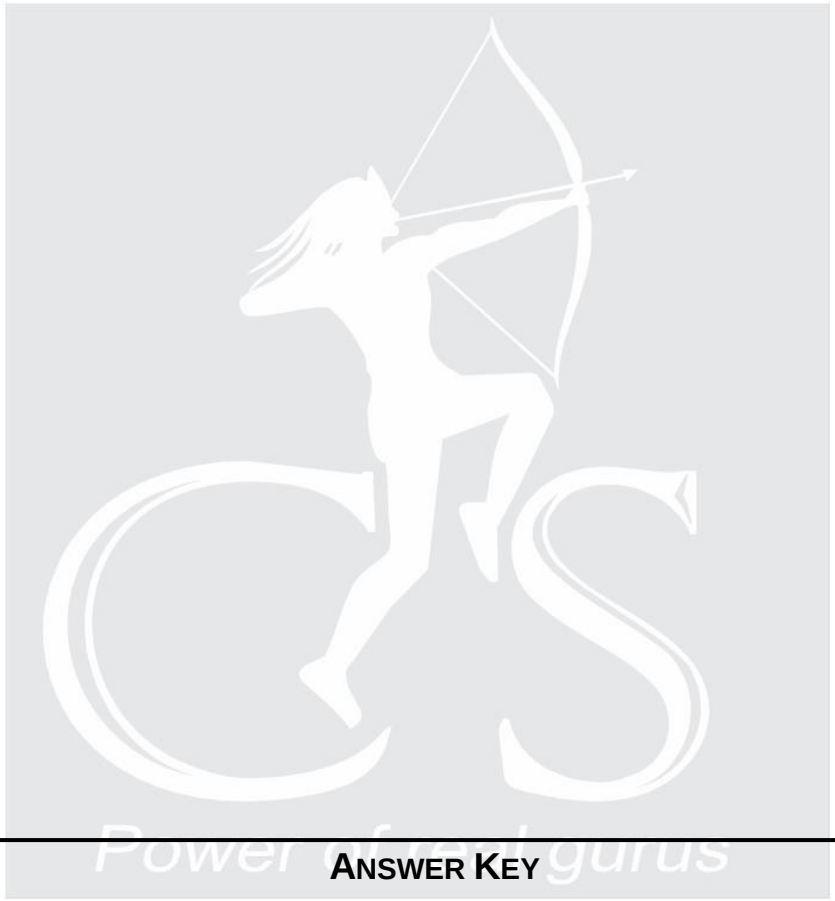
(A) $\rho L/T$ (B) $\sqrt{T/\rho L}$ (C) $T/\rho L$ (D) $2T/\rho L$

24. A capillary tube made of glass of radius 0.15 mm is dipped vertically in a beaker filled with methylene iodide (surface tension = 0.05 Nm^{-1}, density = 667 kg m^{-3}) which rises to height h in the tube. It is observed that the two tangents drawn from liquid-glass interfaces (from opp. sides of the capillary) make an angle of 60° with one another. Then h is close to (g = 10 ms^{-2})

(A) 0.049 m (B) 0.087 m (C) 0.137 m (D) 0.172 m

25. Pressure inside two soap bubbles are 1.01 and 1.02 atmosphere, respectively. The ratio of their volumes is

(A) 2 : 1 (B) 0.8 : 1 (C) 4 : 1 (D) 8 : 1



ANSWER KEY

1.	(D)	**2.**	(A)	**3.**	(D)	**4.**	(C)	**5.**	(C)
6.	(B)	**7.**	(D)	**8.**	(B)	**9.**	(B)	**10.**	(A)
11.	(B)	**12.**	(D)	**13.**	(B)	**14.**	(C)	**15.**	(C)
16.	(C)	**17.**	(C)	**18.**	(A)	**19.**	(D)	**20.**	(A)
21.	(D)	**22.**	(C)	**23.**	(D)	**24.**	(B)	**25.**	(D)